

LIVERNE, MN, USA

WMO: 722006

Lat: 43.621N

Lon: 96.216W

Elev: 436

StdP: 96.19

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54926

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.0	-21.2	-27.7	0.3	-22.8	-25.2	0.4	-21.2	13.4	-4.1	12.0	-6.6	2.4	310	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	32.2	22.4	30.2	21.8	28.7	20.9	24.5	29.5	23.5	28.2	22.5	27.0	5.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.4	27.8	22.2	17.8	27.3	21.1	16.6	26.1	76.8	29.2	72.4	28.4	68.5	26.7	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.6	9.1	DB	-29.4	34.5	3.3	1.4	-31.8	35.5	-33.7	36.4	-35.6	37.2	-37.9	38.2	(4)
(5)			WB	-29.4	26.4	3.2	1.1	-31.7	27.2	-33.6	27.8	-35.4	28.4	-37.8	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-8.2	-7.8	1.1	7.7	14.4	20.3	22.2	20.4	16.6	8.8	1.3	-6.2	(6)	
	DBStd	12.10	7.02	6.40	6.70	5.59	4.91	3.35	2.99	2.88	4.45	5.10	5.66	6.19	(7)	
	HDD10.0	2324	566	499	285	106	16	0	0	0	4	83	264	502	(8)	
	HDD18.3	4239	824	732	536	319	141	18	5	12	85	297	511	760	(9)	
	CDD10.0	1458	0	0	8	38	152	309	377	324	201	47	3	0	(10)	
	CDD18.3	332	0	0	0	1	19	76	124	77	32	3	0	0	(11)	
	CDH23.3	3196	0	0	4	37	253	736	1162	637	325	41	0	0	(12)	
	CDH26.7	911	0	0	0	9	76	222	362	152	82	7	0	0	(13)	
(14) Wind	WSAvg	3.8	4.1	4.1	4.4	4.9	4.6	3.7	2.7	2.4	3.1	3.8	4.0	3.7	(14)	
(15) Precipitation	PrecAvg	690	14	19	36	72	90	111	86	82	73	56	31	20	(15)	
	PrecMax	967	35	49	99	162	172	286	208	170	181	146	102	54	(16)	
	PrecMin	435	3	4	2	13	26	34	10	32	8	11	1	3	(17)	
	PrecStd	122	9	10	23	32	39	50	52	41	49	41	29	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	7.6	22.6	27.7	32.4	33.0	33.0	32.4	31.3	27.5	20.1	10.1	(19)	
		MCWB	3.9	4.0	12.8	15.0	17.5	20.5	24.7	24.0	21.1	17.0	12.4	6.9	(20)	
	2%	DB	3.0	4.9	17.8	22.9	28.0	31.2	32.2	30.1	29.0	23.0	16.1	6.0	(21)	
		MCWB	1.3	2.2	11.5	12.9	17.3	20.6	24.1	23.3	20.4	14.7	9.7	2.9	(22)	
	5%	DB	2.3	2.5	13.7	20.2	26.2	29.0	30.2	27.8	27.2	21.0	12.4	2.8	(23)	
		MCWB	0.9	1.0	9.7	11.6	16.7	20.0	23.1	21.5	18.8	13.6	7.8	1.5	(24)	
	10%	DB	1.1	1.1	10.3	17.3	23.0	27.4	28.0	27.1	25.0	17.5	9.9	2.0	(25)	
		MCWB	0.3	0.1	7.5	10.1	14.7	19.0	21.4	20.9	18.0	11.6	6.3	1.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.7	15.2	16.4	21.4	24.1	26.7	25.5	23.3	19.6	13.7	7.2	(27)	
		MCDB	7.2	7.5	20.1	22.8	26.9	30.0	31.6	30.1	28.3	23.8	18.3	9.8	(28)	
	2%	WB	1.8	2.5	12.3	14.4	19.2	22.8	25.1	24.0	21.8	16.5	10.5	3.2	(29)	
		MCDB	3.4	4.4	16.2	21.0	25.7	28.4	30.2	28.6	26.3	21.1	15.1	5.2	(30)	
	5%	WB	0.8	1.1	10.0	12.9	17.7	21.6	23.9	22.9	20.6	14.2	8.1	1.8	(31)	
		MCDB	1.7	2.3	14.5	18.2	23.5	26.9	28.8	27.0	25.4	19.4	11.9	3.1	(32)	
	10%	WB	0.1	0.0	7.1	11.4	16.4	20.4	22.9	21.9	19.1	12.3	6.1	0.7	(33)	
		MCDB	0.7	0.7	10.3	16.1	21.9	25.8	27.5	25.8	23.7	16.5	9.9	1.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.6	9.6	11.5	12.0	11.3	11.6	11.3	13.0	11.9	10.7	8.8	(35)	
		MCDBR	10.3	10.1	16.1	17.4	15.7	13.8	12.5	12.5	15.1	16.9	15.5	10.1	(36)	
	5% WB	MCWBR	8.8	8.3	9.9	8.9	7.2	6.2	6.6	6.9	7.2	9.0	9.7	8.1	(37)	
		MCWBR	9.2	9.6	15.2	14.5	12.8	11.3	11.5	11.2	12.4	13.8	14.4	9.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.267	0.282	0.315	0.370	0.390	0.400	0.408	0.411	0.361	0.325	0.290	0.274	(40)	
		taud	2.437	2.416	2.415	2.288	2.308	2.339	2.309	2.277	2.394	2.472	2.491	2.436	(41)	
	Edn at Noon		878	922	928	894	882	872	859	844	864	858	842	831	(42)	
		Edh at Noon	72	88	102	125	127	123	126	126	103	84	69	66	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.90	2.96	3.86	4.83	5.56	6.23	6.63	5.48	4.37	2.96	2.03	1.51	(44)	
	RadStd		0.18	0.24	0.31	0.44	0.41	0.40	0.25	0.41	0.32	0.36	0.14	0.13	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page